

Skill Arena - Phase 5 Document 4

Database Architecture, Caching Strategy, Data Storage, Replay Storage & High Availability Framework

Document Purpose

Define how data is stored, cached, replicated, protected and made highly available across the platform.

Business Objective

Provide reliable, scalable and fault-tolerant storage for gameplay, financial, replay and operational data.

Architecture Philosophy

Separate data by domain ownership, maximize resilience and ensure long-term auditability.

Database Strategy

Domain-driven databases assigned to User, Gameplay, Wallet, Treasury, Tournament, Replay and Security services.

Primary Data Domains

Identity Data, Gameplay Data, Financial Data, Tournament Data, Replay Data, Compliance Data and Security Data.

Database Ownership

Each microservice owns its own authoritative datastore.

Read/Write Separation

Support dedicated read replicas and optimized reporting workloads.

Data Replication

Multi-zone and multi-region replication for resilience and disaster recovery.

High Availability Design

Automatic failover, redundant infrastructure and continuous health monitoring.

Replay Storage Architecture

Store replay metadata separately from replay event streams and replay assets.

Replay Data Components

Replay IDs, Validation Hashes, Event Streams, Match References and Integrity Records.

Long-Term Storage

Archive historical replays, audit records and financial history for compliance and investigations.

Caching Strategy

Use distributed caching for rankings, profiles, sessions, leaderboards and frequently requested data.

Cache Invalidation

Event-driven invalidation and expiration policies to maintain consistency.

Leaderboard Storage

Optimized storage and caching for seasonal, lifetime and tournament rankings.

Analytics Storage

Dedicated analytical storage for reporting, forecasting and business intelligence.

Backup Architecture

Automated backups, point-in-time recovery and integrity verification.

Disaster Recovery

Recovery procedures, regional redundancy and business continuity support.

Data Retention Policy

Permanent retention for financial and audit records. Policy-based retention for operational data.

Encryption Requirements

Encryption at rest and in transit for all sensitive information.

Access Controls

Strict role-based access controls for databases, storage systems and archives.

Monitoring Framework

Database performance, storage growth, replication health and backup monitoring.

Scalability Requirements

Support billions of records and millions of concurrent users.

Performance Requirements

Low-latency reads, efficient writes and predictable scaling characteristics.

Audit Requirements

Database changes, access events and storage operations permanently logged.

Acceptance Criteria

Platform data remains secure, available, recoverable and scalable under global workloads.